GJESUS3 RESEARCH DATA MANAGEMENT · NUCLEAR IMAGING

NI Live Sync

Getting Molecubes PET/SPECT/CT scans from the acquisition Mac into gjesus3, registered, checksummed and filed in the right project, without researchers carrying files by hand.

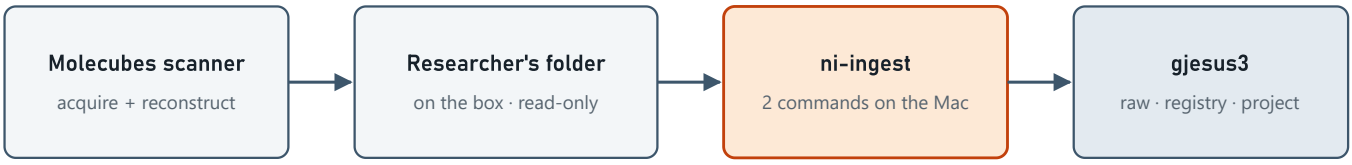
● **Proven in production · ready for a supervised pilot · merge into main pending**

95/95
reconstructions in the first production sync (Irene, 8 Oct), 0 failed

75
already-loaded reconstructions recognised and skipped: no duplicates

52
CT attenuation maps filed as derived files, not raw

1,945
NI acquisitions in gjesus3 production today, across all routes



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Prepared for	Ryan Tasseff, Data Management Lead, and the MFB data team
Prepared by	Data Office, gjesus3 RDM (feat/ni-live-hardening)
Date	9 October 2026
Status of record	tasks/STATUS.md · tasks/RESUME_ni_live.md · CHANGELOG.md

1 · EXECUTIVE SUMMARY

What it is, where it stands, what we need from you

The problem

NI researchers run the Molecubes scanner themselves. Getting data out has meant staying in the acquisition room and copying reconstructions by hand to the `gnuclear` working space. There the same reconstruction ends up in many folders (up to 48 copies of one scan), filed by personal habit. None of it is registered, checksummed or tied to its animals and project. When the August backfill pulled that space into gjesus3, it found 2,312 acquisitions, and 673 of them still cannot be filed because no protocol code was recorded.

The solution

A sync the researcher runs at the Mac when reconstructions finish. It is two commands: `--plan` shows what's new, then a second run syncs it. Each reconstruction is copied to gjesus3 `raw/`, checksummed, described in a metadata sidecar, registered and filed in its project. The researcher's folder on the box is only ever read. Mistyped folder names and the tracer are fixed once, in a small per-researcher corrections file.

Where we are

- **Proven in production on 8 Oct.** Irene's whole box folder: 95/95 reconstructions (6.07 GB, 25 min, Mac load ≤1.7). Every check passes: checksums 95/95, registry validator 0 errors, 95 project links as true hard links, and 145 animal records matching the animal-facility DB.
- **Access solved.** Since 1 Oct the Mac is operable from the Data Office through a hardened reverse SSH tunnel. Unai (NI platform manager) approved remote sync tests on 2 Oct.
- **Two serious defects caught before they reached researchers**, both fixed and tested: re-ingesting scans another route had already loaded (75 would have been duplicated), and a recovery tool writing one mouse's record into another's block (25 sidecars, repaired and verified the same night).
- **Ready for researchers:** a production copy on gnuclear, an `ni-ingest` shortcut on the Mac, and an illustrated guide. 47 automated test suites pass, and the branch merges cleanly into main.

Decisions needed (section 9)

- Merge into main, timed with the coordinator session.
- A date for Irene's first supervised sync.
- Whether to recover 292 older non-NI animal records (103 already match the DB).
- A go for the clean-up phase.
- Re-confirm the interim use of your gjesus3 login on the Mac.

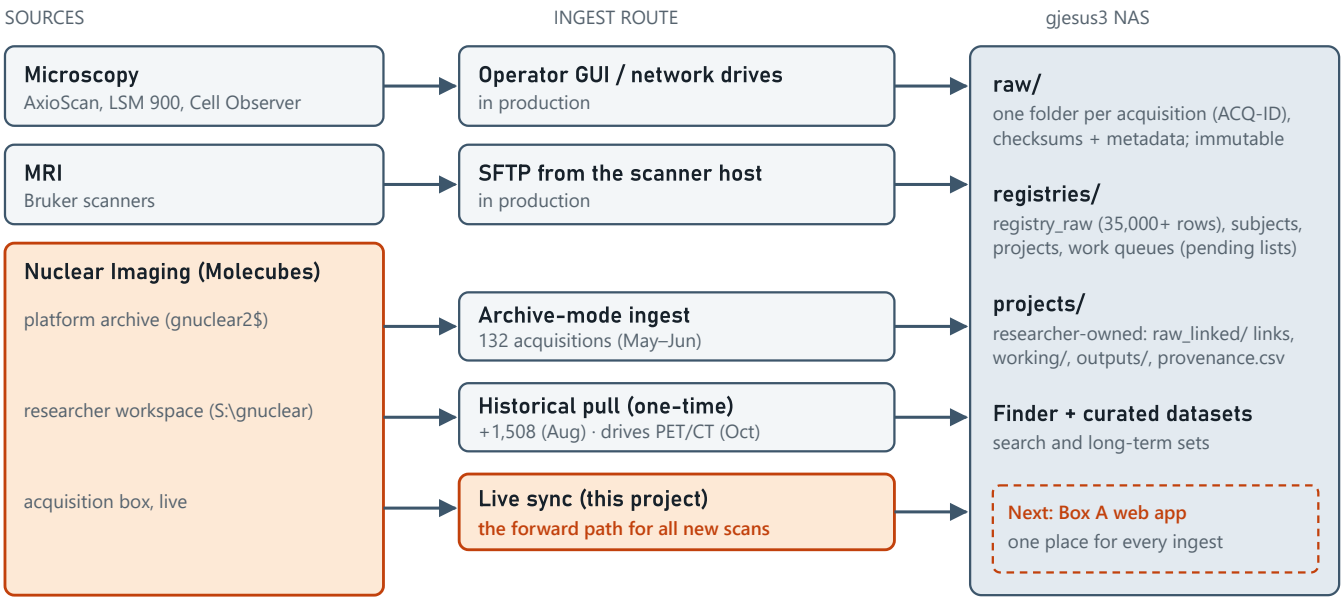
Next two weeks

- Merge; restage the production copy from main.
- Irene syncs herself, supervised; then the next researcher.
- A read-only `--plan` of each researcher's folder before they start.
- The Data Office follow-up routine after each sync (links + animal records).
- Clean-up: archive old notes and test kits; keep every working instruction.

2 · CONTEXT

How the sync fits into gjesus3

gjesus3 is the research-facing working layer: an immutable `raw/` store where every acquisition gets an ACQ-ID, registries that are the system of record, and researcher-owned project folders that link to the raw data.

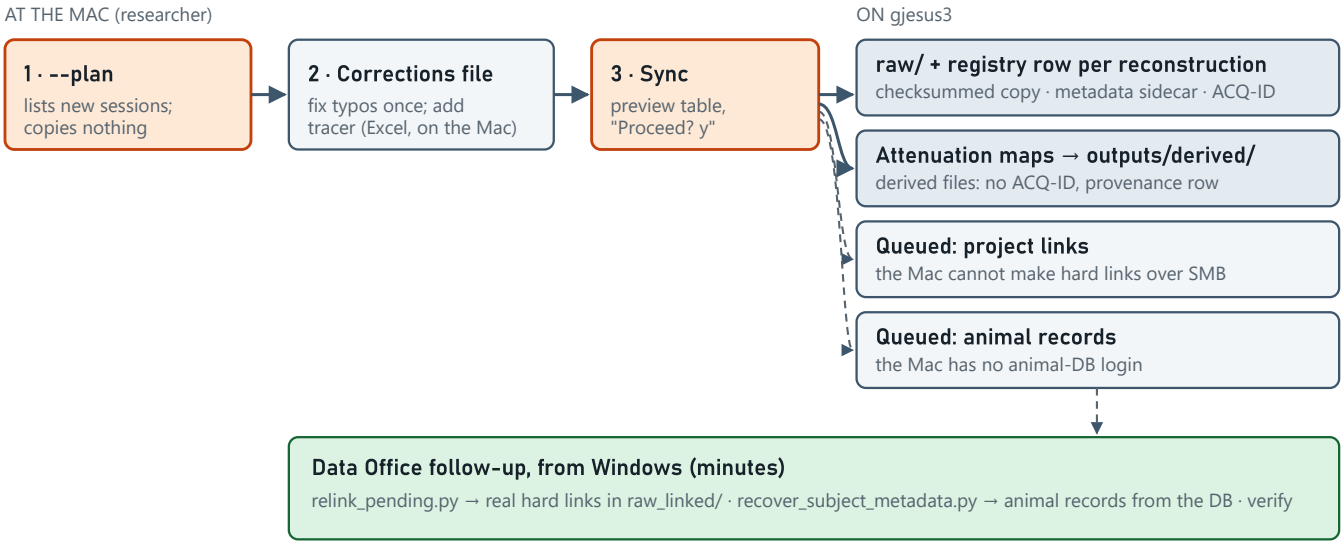


NI has three routes into gjesus3. The archive and the historical pull filled the past. The live sync is how new scans arrive from now on, and it recognises what the other two already loaded.

NI ROUTE	WHAT IT COVERS	IN PRODUCTION	STATUS
Archive-mode ingest	Platform's standardised archive (`gnuclear2$`)	132 acquisitions	DONE
Historical pull	Researchers' working copies on `S:\gnuclear`, 2022–2026; historical drives	1,718 acquisitions (incl. drive PET/CT)	DONE 673 held back for protocol codes
Live sync	New scans, straight from the acquisition box	95 (first run, Irene)	PILOT

3 · WHAT WE BUILT

The sync, end to end



What a researcher does (left) and what lands on gjesus3 (right). The two queued parts are finished from the Data Office workstation, where hard links and the animal database are available.

Design principles

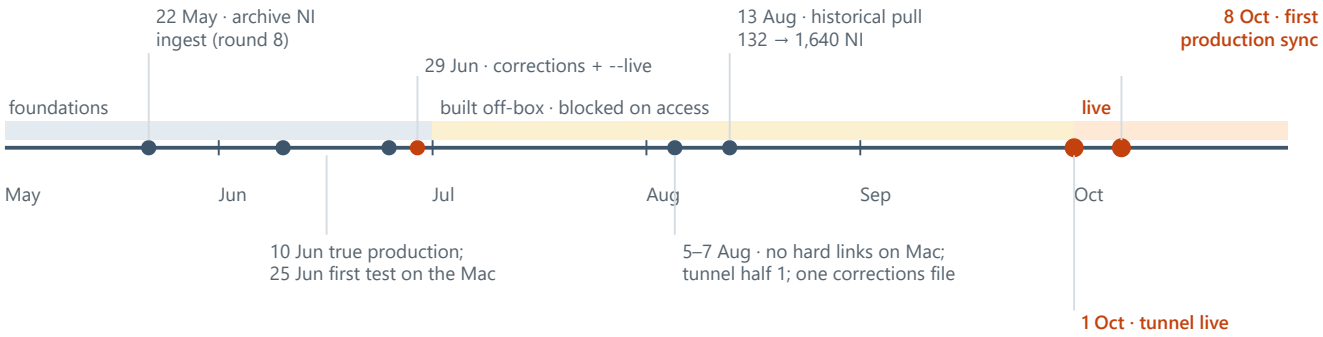
- **The box is read-only.** The sync never moves, renames or deletes anything in a researcher's folder.
- **One acquisition per reconstruction.** A reconstruction that finishes weeks later becomes its own entry and inherits the session's corrections.
- **Fix once.** A per-researcher corrections file holds typo fixes and the tracer, permanently.
- **Nothing twice.** Scans loaded by another route are recognised by timestamp, modality and reconstruction number.
- **Derived is not raw.** CT attenuation maps go to the project as derived files, traceable to their scan.

Rules that shaped it

- **The platform's box, the platform's terms.** Small CPU and I/O footprint, nothing written outside one Data Office folder, nothing deleted. Unai's concern is code nobody can read and anything that erases.
- **Demonstrable, not just readable:** `--plan` and `--dry-run` show every path read and written, and touch nothing.
- **Project names** (your rule, 8 Oct): 4 digits in the code position = `AE-biomaGUNE-NNNN`; anything else = `Project-<code>`. No DB check on the Mac; the later animal lookup surfaces bad codes.
- **No questions mid-sync:** one batch-wide answer can't be right for many studies.

4 · HISTORY

Five months, three phases

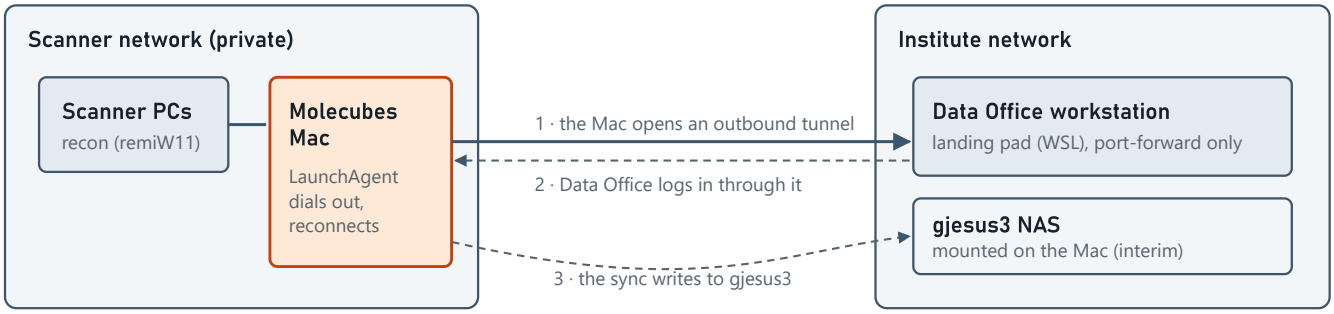


WHEN	MILESTONE	WHY IT MATTERED
22–27 May	Archive-mode NI ingest; slim layout keeps only reconstructed DICOMs	Avoided copying ~358 GB of platform raw
12–29 Jun	Live discovery grammar; first test on the Mac (into a sandbox); one acquisition per reconstruction; corrections + tracer; `--live` mode	The design that is in production now
29 Jun → 6 Aug	Two synthetic test acquisitions slipped into production through a shell setting; found in review, fully reversed	Every test since pins a scratch NAS
5–7 Aug	On the box: macOS cannot make hard links over SMB → links are queued and made from Windows. Tunnel workstation half built. Corrections become one file per researcher	Removed the last design blockers off-box
13 Aug	Historical pull from `S:\gnuclear`: +1,508; review stopped 21 fabricated projects	The past is loaded; the live sync must not duplicate it
1–2 Oct	Tunnel's box half installed on a site visit; Data Office can operate the Mac; Unai OKs remote tests	Ended a month without access
6 Oct	Caught up with main (303 commits); end-to-end test harness; all four merge gates pass on the Mac against a scratch NAS	Engine proven on the real box
7–8 Oct	gjesus3 kept mounted on the Mac; your hands-on test; real-folder dry runs find duplicates and attenuation maps; both fixed; **first production sync**; links + animal records; guide and production copy	Ready for a supervised pilot

5 · ACCESS TO THE BOX

The reverse tunnel, and the rules for using it

The Molecubes Mac sits on the scanner's private network, so nobody could reach it from outside. Every test before October needed a person in the acquisition room, and access was lost for a month.



Security posture

The Mac's tunnel key is restricted to port-forwarding; it cannot run commands on the workstation. The Data Office's login key to the Mac is recorded, and removable in one step.

Your rules for the Mac

Write freely only in your gnuclear folder. Every other write needs your OK. Never delete what we didn't write. Keep reads small and runs at low priority, outside acquisitions.

Known limits

After a Mac reboot the tunnel returns only when someone logs in (no auto-login). The gjesus3 mount uses your login for now (section 8); MacMounter re-mounts it automatically within 2 minutes.

> Box A, the planned server, takes over the tunnel without a site visit. The move is decided and planned make-before-break: a second connection is proven before the first is removed. Once Box A pulls data through the tunnel, the Mac needs no gjesus3 mount and no stored credentials.

6 · RESULTS

The first production sync, 8 October

Irene's whole box folder, run from the Data Office through the tunnel, at lowest priority, outside acquisitions. Before it, a read-only plan and two dry runs. After it, independent verification.

From folder to production

222 reconstructions in Irene's folder that the sync had never seen (920 folders scanned, 50 sessions)

75 already in production → skipped

52 attenuation maps

→ projects' outputs/derived/, with provenance, no ACQ-ID

95 reconstructions ingested (6.07 GB)

→ AE-biomaGUNE-1025 ×54 · 0522 ×33 · 1125 ×8
no project created; 5 typo sessions corrected

Same night, from Windows

95 project hard links created · 95 animal records filled from the animal-facility DB (57 animals, 145 subject blocks)

VERIFICATION (READ-ONLY, FROM THE WORKSTATION)	RESULT
Registry rows added = live-sync rows, routed as previewed	95 / 95
Checksums re-verified on gjesus3	95 / 95
Registry validator, all of production	0 ERRORS
Project links are true hard links (same file as raw)	95 / 95
Attenuation maps placed, each with a provenance row tied to real ACQ-IDs	52 / 52
Animal records equal the DB record of their own animal	145 / 145

Footprint on the platform's Mac

- Sync: 25 minutes for 95 reconstructions; system load stayed ≤1.7, and the scanner software was idle.
- Read-only plan of the whole folder: 70–90 seconds.
- Nothing written on the box's data; every test write landed in one Data Office folder, proven by before/after snapshots.

Typo caught by the process

Five sessions were typed `1015_m…` under series `1025`, and the folder date was a year off. Irene's own gnuclear copy had the same animals under 1025, so they were corrected in her corrections file before the sync.

7 · ISSUES

What went wrong, and what we did about it

Most of these were found by the project's own checks before they could affect researchers. One wrote bad data into production: it was found and repaired within the hour, and is reported in full.

ISSUE	IMPACT	RESOLUTION	STATUS
No access to the box	A month without access; every test needed a visit	Reverse SSH tunnel, hardened; remote tests approved by the platform	RESOLVED
macOS cannot hard-link over SMB	Project links can't be made from the Mac	Links are queued and made from Windows in minutes (95 done 8 Oct)	BY DESIGN
No animal-DB login on the Mac	Animal details missing at sync time	Queued and filled from the workstation; the recovery tool was fixed (below)	BY DESIGN
gjesus3 mount kept dropping	The sync's destination unavailable	Cause: no saved password. Interim: your login, kept up by MacMounter; a race between its three copies was found and fixed	INTERIM
Duplicates across routes	28 of 54 sessions were already loaded by the August pull; 75 reconstructions would have been copied again	Reused the existing cross-route guard, per reconstruction; verified on real data	FIXED
Attenuation maps looked like reconstructions	52 derived files would have become raw acquisitions	Your ruling: derived. They go to the project's `outputs/derived/` with provenance	FIXED
Batch-wide question mid-sync	One `is_control` answer applied to many studies	Found in your hands-on test; live mode never asks; a test now simulates a person at a terminal	FIXED
Recovery wrote the wrong animal	25 two-mouse sidecars got the second mouse's sex, date of birth and procedures	Backed up, confirmed, reset and re-recovered with a fixed per-animal tool. All 145 blocks verified against the DB. It never reached the subjects table	REPAIRED
Recovery couldn't find sidecars from Windows	No animal record had ever been recovered from Windows	Paths are now resolved under the NAS root; 292 older non-NI rows await your call	YOUR CALL
Corrections file read-only from Windows	Researchers can't fix codes from their desks	Edit on the Mac (Excel is there); a web form on Box A later	ACCEPTED
Folder-name typos invent projects	e.g. `1015` would create a fake project	Your naming rule, and the later DB lookup flags bad codes; corrections fix them before syncing	MANAGED
Synthetic test data in production (June)	2 test acquisitions in true production	Fully reversed on 6 Aug; every test since pins a scratch NAS	RESOLVED

Lesson recorded: a clean dry run of an older tool is not proof that it suits new data. Before applying a repair tool to production, check how it handles this data's shapes (several animals per scan), back up what it touches, and verify every written record against an independent source.

8 · RISKS AND OPEN ITEMS

What could still bite, and the mitigation in place

RISK	LIKELIHOOD	IMPACT	MITIGATION / PLAN
Your gjesus3 login is used on a shared Mac account (superuser rights, password in its keychain)	MEDIUM	HIGH	Mount hidden from Finder; QNAP snapshots allow recovery from deletions. IT declined a dedicated account. Retired when Box A pulls through the tunnel; undo script ready.
A researcher sync overlaps a Data Office batch (both write the registry)	MEDIUM	MEDIUM	Files stay consistent (locking), but batch checks stop. Interim: researchers sync outside announced windows. A "batch running" flag is in the BACKLOG at moderate priority.
Folder-name mistakes in other researchers' trees	HIGH	LOW	Naming rule + corrections file; a read-only `--plan` of each researcher's folder before onboarding; DB lookup flags bad codes later.
Follow-ups (links, animal records) forgotten after researcher syncs	MEDIUM	LOW	Data is safe either way; a manual routine for now; a scheduled checker for all three work queues was designed in August but not built, and was not in the BACKLOG as this report first said (corrected 2026-11-03).
Losing the platform's trust (footprint, unreadable code, anything that erases)	LOW	HIGH	Your rules for the Mac, low-priority runs, read-only on the box, approved remote tests. Long-term the Mac runs nothing (Box A).
Mac reboot leaves the tunnel down until someone logs in	LOW	LOW	Researchers log in to scan anyway; the field card covers recovery.
Two folders named `new recon` are not parsed (Irene)	KNOWN	LOW	Reported in every sync summary; BACKLOG item.

Parked by design (BACKLOG)

- Researcher name casing (`Irene` / `irene`): the group decides the convention first.
- Per-session `is_control`: a field on the Box A corrections form.
- `session_id` semantics (ISA study level): a design review.
- 673 historical NI acquisitions waiting for protocol codes.

Clean-up phase (after the pilot)

- Keep every working instruction (Box A repeats the setup): field card, tunnel notes, mount kit.
- Archive superseded plans; shorten the RESUME to a current-state page.
- Remove test kits and scratch NAS copies from the gnuclear Data Office folder.

9 · WHERE WE GO NEXT

Roadmap and decisions

NOW · this week

- Merge into main (with coordinator)
- Restage production copy from main
- Irene's first supervised sync
- Follow-ups: links + animal records
- Share the researcher guide

NEXT · Oct–Nov

- Onboard researchers one by one (read-only plan of each folder first)
- Data Office follow-up routine
- Clean-up phase
- Batch-window flag (moderate)
- 292 older animal records (if approved)

LATER · Box A

- Port gjesus3 production to Box A
- Move the tunnel, no site visit
- One ingest web app for all instruments
- Box A pulls NI scans: researchers leave the room when the scan ends
- Corrections as a web form; Mac runs nothing

DECISION FOR YOU	OPTIONS	RECOMMENDATION
Merge into main	Now (merges cleanly) vs after the pilot	**Now, timed with the coordinator. Merging deploys nothing; researchers run the staged copy.**
Irene's supervised first sync	Pick a slot after the coordinator's MRI work	**Early next week, at the Mac with the guide, Data Office watching through the tunnel.**
292 older non-NI pending animal records (103 match the DB)	Recover now with the fixed tool / leave	**Recover, scoped and verified like the NI 95; check multi-animal cases first.**
Clean-up phase	Go after the pilot / now	**After the pilot, so nothing is removed that the pilot still needs.**
Interim credentials on the Mac	Keep until Box A / revisit with IT	**Keep, with the risk noted. Ask IT again before onboarding all researchers.**

Decided in this cycle (no action needed): the cross-route duplicate guard, attenuation maps as derived files, the project-naming rule, and the batch-window flag as a moderate BACKLOG item.

APPENDIX

Terms and where things live

TERM	MEANING
Acquisition / ACQ-ID	One registered unit in `raw/`; for NI, one reconstruction. ID like `ACQ-20260522-CT-002`.
Reconstruction	An image the scanner software computes from a scan (`recon_<n>/`); a scan can have several, added later.
Session	One animal (or group) in one sitting: `series/date/subject` on the box.
Sidecar	`metadata.json` next to each acquisition: scan, animal and processing details.
Corrections file	Per-researcher CSV on gnuclear: typo fixes and the tracer, applied by every sync.
Derived file	Made from an acquisition (e.g. a CT attenuation map); lives in the project, not `raw/`.
Work queues	Pending lists for links and animal records, finished from Windows.
Tunnel	Outbound SSH from the Mac that lets the Data Office operate it.

WHAT	WHERE
Researchers' copy of the sync	`S:\gnuclear\2026\Jesus\_gjesus3_sync\`
Researcher guide	`…\_gjesus3_sync\NI_SYNC_GUIDE.html`
Command on the Mac	`ni-ingest <folder> --plan`, then `ni-ingest <folder>`
Code + history	branch `feat/ni-live-hardening` (pushed), merging into `main`
Current state, next steps	`tasks/RESUME_ni_live.md`, `tasks/STATUS.md`
Tunnel record	`equipment/nuclear-imaging/live_machine_remote_access.md`
Sync rules + layout	`equipment/nuclear-imaging/live_machine_data_layout_and_sync_rules.md`
Data Office runbook	`tools/operator/NI_LIVE_RUNBOOK.md`
Later work	`tasks/BACKLOG.md` (Box A, web app, flag, …)

People

Ryan Tasseff, Data Management Lead (owner) · **Unai**, NI platform manager (owns the acquisition Mac; approved remote tests) · **Irene**, NI researcher (first production sync, pilot user) · **Coordinator session**, sequences registry writes across the Data Office's parallel work.